

How coding assistants actually differ.

A reproducible field study of assistant products, frontier models, open-weight workers, neutral harnesses, and distributed AI development.

01

PRINCIPAL FINDING

**The strongest setup is a portfolio,
not a winner.**

Codex leads. Claude challenges. OpenCode routes bounded workers. Pi keeps the model comparison honest.

Coding Intelligence Field Study

3 ASSISTANTS / 5 NEUTRAL MODEL ROUTES / 20 PHASE 2 OBSERVATIONS

00 / ABSTRACT

Separate the product from the model.

Phase 1 measured complete assistant experiences. Phase 2 held the Pi harness constant and changed only the model route. Together, the two phases distinguish product advantage, model capability, and workflow design.

RESEARCH CLAIM

A strong AI development practice should combine specialized roles rather than choose one permanent winner: a frontier planner and integrator, an independent reviewer, economical bounded workers, and a neutral model harness.

3

primary coding assistants

2

completed research phases

20

neutral Phase 2
observations

5

models in the fixed Pi
harness

RESEARCH QUESTIONS

Q1

Which assistant should lead?

Codex was the strongest default integrator; Claude was the strongest independent reviewer.

Q2

Is OpenCode worth exploring?

Yes, when model flexibility and bounded open-model work matter more than native polish.

Q3

Can economical models implement?

Yes. Qwen and GLM passed hard tasks when work was explicit and verifiable.

Q4

Should every task be distributed?

No. Distribution improved separation and auditability but was 3.57x slower in the measured route.

01 / INTRODUCTION

The object of study is the whole coding system.

A coding result reflects a model, an assistant product, a harness, a permission boundary, and a verification policy. This study makes those layers visible enough to support an engineering decision.

PROBLEM

A model leaderboard cannot explain whether a failure came from reasoning, tool use, permissions, orchestration, or the task contract when all of those variables change together.

RESEARCH QUESTIONS

Q1

Which assistant products are best suited to lead, implement, review, and integrate?

Q2

How much of an observed result belongs to the model versus the assistant and harness?

Q3

When does distribution improve evidence or risk control enough to justify its cost?

Q4

Can an open-weight-only route complete a high-risk task with independent acceptance?

CONTRIBUTIONS

Two-phase product/model separation; frozen contracts and private evaluators; role-level operating guidance; and an open-weight authorization trial that preserves both success and evidence failure.

02 / RELATED WORK

Positioning this study among coding-agent evaluations.

The study borrows the realism, hidden-test discipline, and tool-mediated framing of established benchmarks while changing the unit of analysis to the product-model-workflow combination.

01

SWE-bench[Real GitHub issues](#)

Issue-resolution evaluation on real repositories.

02

SWE-Lancer[Software-development work](#)

Broadens the surface beyond isolated bug fixes.

03

Terminal-Bench[Tool-mediated terminal tasks](#)

Measures agents operating through shell tools.

04

RE-Bench[Long-horizon engineering](#)

Studies research and engineering trajectories.

05

UTBoost[Evaluator integrity](#)

Shows how weak tests can overstate patch correctness.

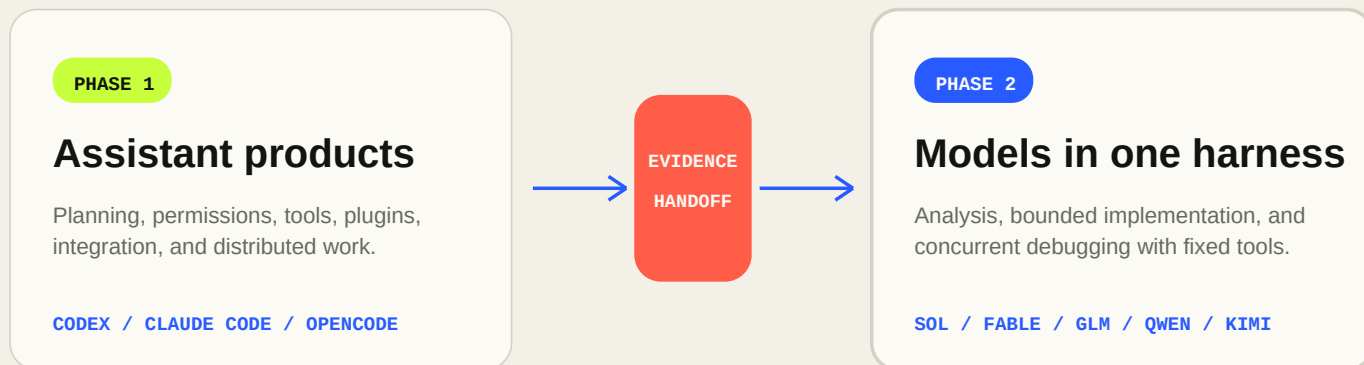
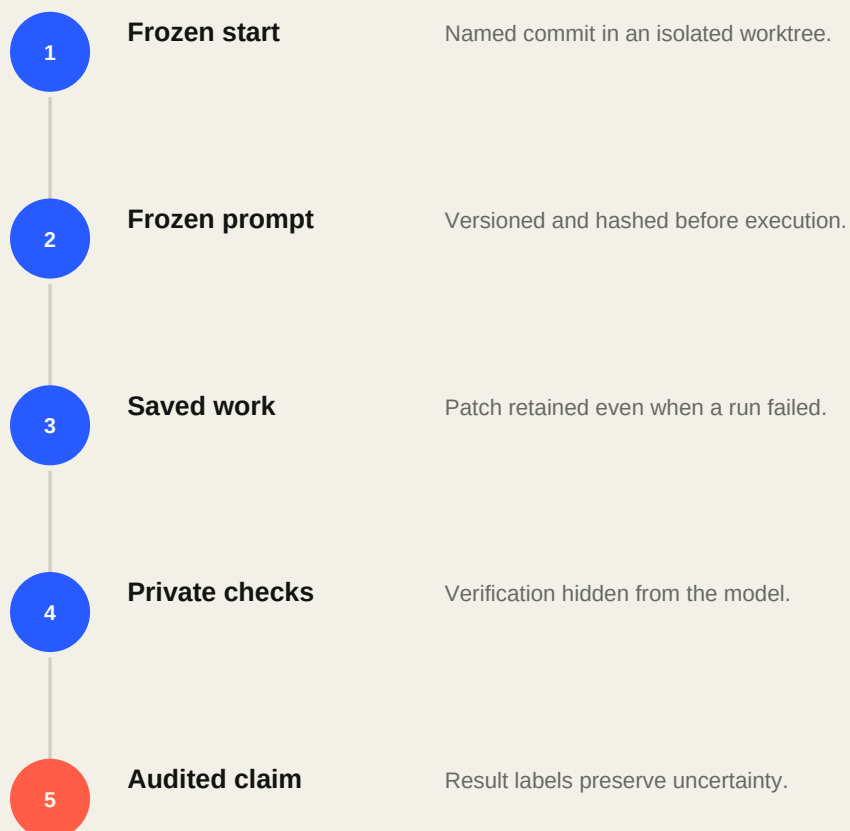
POSITIONING

This is an applied field study, not a replacement leaderboard. Its contribution is attribution: separating product behavior, model behavior, orchestration cost, and evidence quality on versioned synthetic tasks.

03 / STUDY DESIGN

Two phases, one evidence chain.

The design first tested whole assistant products, then isolated model behavior inside a fixed control surface.

**EVIDENCE CHAIN**

04 / THE ASSISTANTS

Four tools. Four different jobs.

The assistant is the system around the model: tools, permissions, context, orchestration, and developer experience.

LEAD AND INTEGRATOR

01

Codex**Best default lead**

Plan, implement, integrate, and run the final gate.

REVIEWER AND ALTERNATE LEAD

02

Claude Code**Strongest challenger**

Challenge plans and perform fresh high-risk reviews.

OPEN-MODEL WORKER

03

OpenCode**Worth keeping**

Route hosted open models to narrow, independently testable tasks.

NEUTRAL MODEL HARNESS

04

Pi**The laboratory**

Compare models without mixing in assistant-product differences.

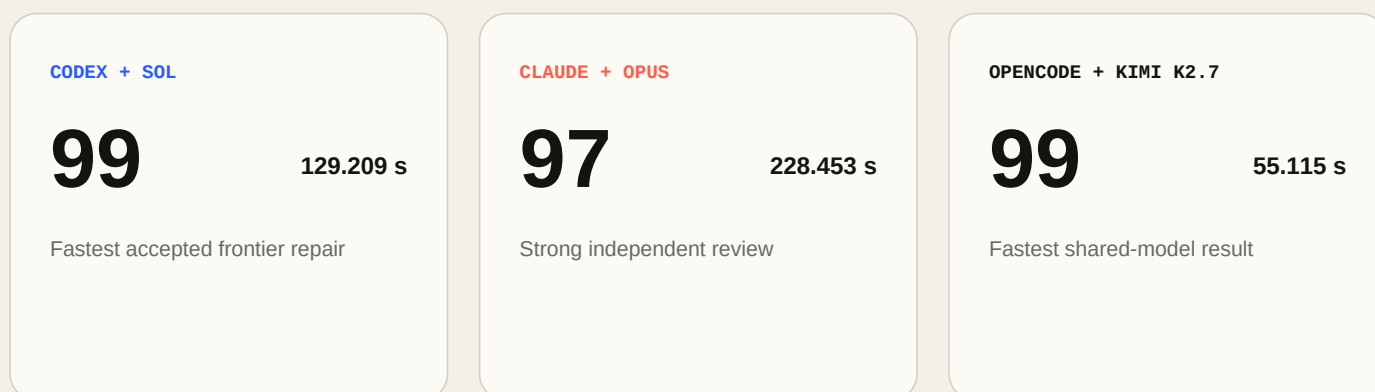
OBSERVED OPERATING RULE

Use Codex to lead, Claude to challenge, OpenCode to route bounded open-model work, and Pi to compare models neutrally.

05 / PHASE 1

Products under development pressure.

The assistant study moved from repository analysis to fixes, interfaces, an incident, cross-assistant tooling, and a securitized-products trade-capture system.



MEASURED DISTRIBUTED WORKFLOW

Correct and auditable - but 3.57x slower.

Codex planned, Claude challenged, OpenCode implemented with Kimi, Codex integrated, and Claude reviewed. The route scored 100 with zero frontier repair.



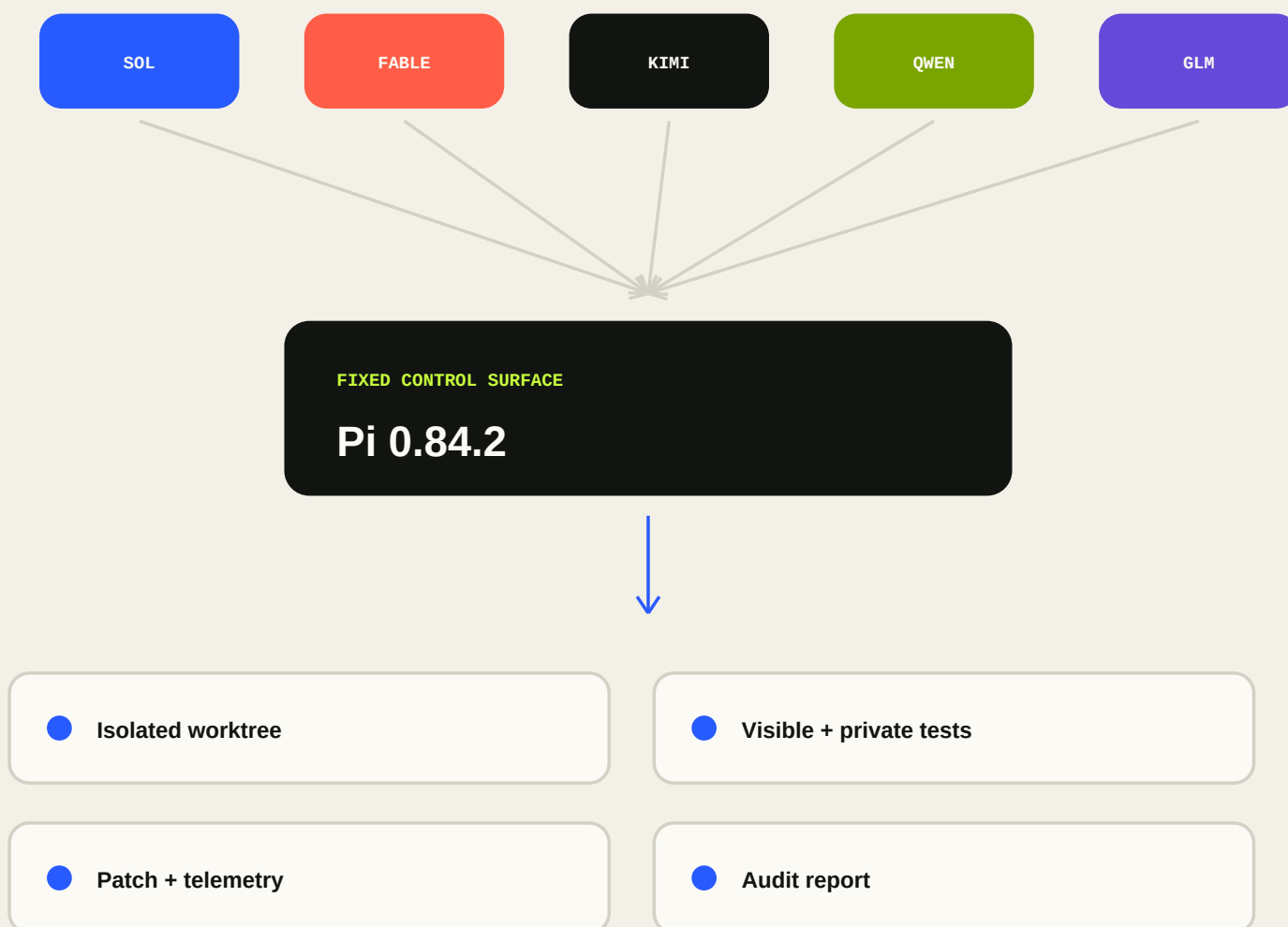
DECISION

Do not distribute every task. Use the multi-assistant route when independent challenge, ownership separation, or auditability justifies the coordination cost.

06 / PHASE 2 METHOD

Five models. One neutral Pi harness.

The baseline, prompts, tools, provider, timeouts, and zero-intervention policy stayed fixed. Only the model route changed across 20 counted observations.



CONTROL RULES

- Fresh session for every route
- Fixed timeout and zero intervention
- Private verification preserved
- Early-edit requirement
- Saved patch after failure
- Subscription, API, and provider costs labeled separately

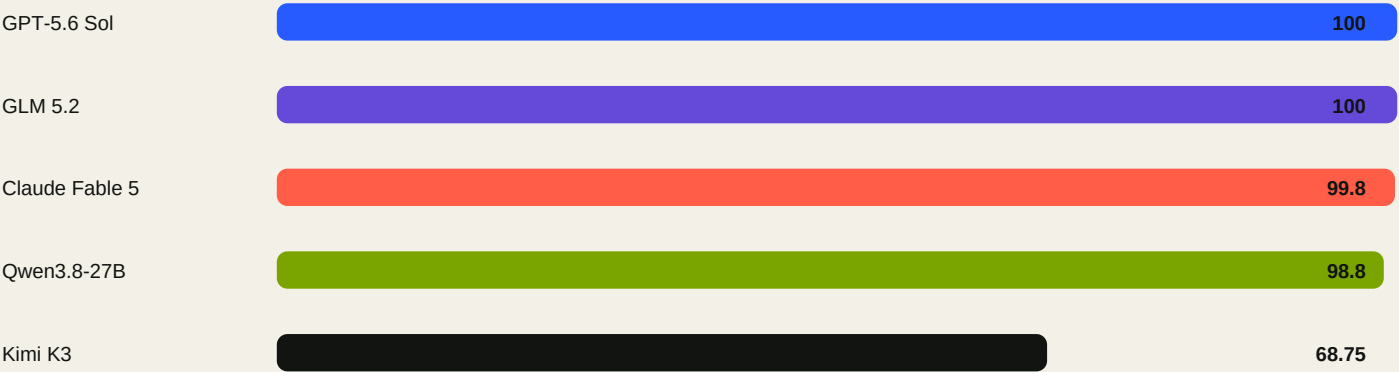
Capability clustered. Reliability separated.

Sol and GLM tied at 100. Qwen approached frontier quality at the lowest saved Pi cost. Kimi remains penalized for a counted timeout even though its saved patch later scored 95.

STRICT WEIGHTED COMPARISON

MODEL	SCORE	WRITE	SECONDS	PI COST
GPT-5.6 Sol	100	3/3	333.389	\$0.0808
GLM 5.2	100	3/3	744.357	\$0.0562
Claude Fable 5	99.8	3/3	241.862	\$0.3644
Qwen3.8-27B	98.8	3/3	621.744	\$0.0160
Kimi K3	68.75	1/3	1178.618	\$0.0760

STRICT SCORE



COST RECONCILIATION

Saved Pi telemetry: \$0.5933

Observed account decrease: about \$2.50

The \$1.9067 difference cannot be allocated by model; route-level cost rankings remain provisional.

08 / OPEN-WEIGHT TRIAL

A bounded authorization task under an independent gate.

The new trial kept the task, tools, and acceptance policy fixed while routing planning, implementation, review, and audit through open-weight models.

RECORDED ROUTE

GLM 5.3 -> Qwen3-Coder-Next -> GLM 5.3 Flash

Plan -> implement -> review -> final audit on AR8, a synthetic multi-tenant authorization task.

2/10

independent baseline

10/10

external evaluator

\$0.46

metered spend

0/18

candidate suite

ROLE HANDOFF**INTERPRETATION**

The route cleared the independent acceptance gate and produced a usable patch at low metered cost. The candidate-authored regression suite reported 0/18 because it referenced a nonexistent createRecord API; that evidence failure is retained rather than hidden.

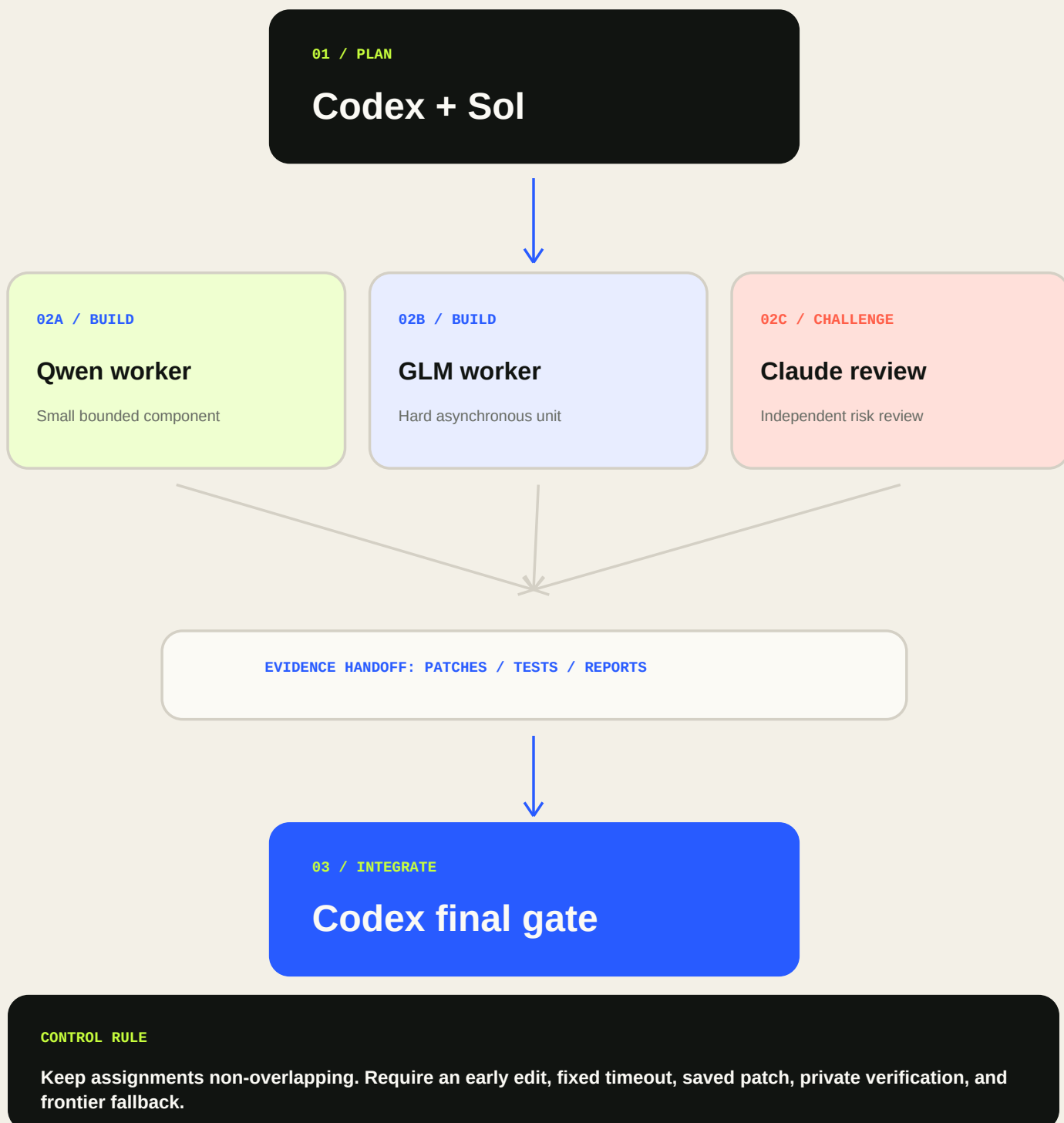
CLAIM BOUNDARY

One successful open-weight observation supports a route hypothesis, not a universal ranking or production security certification.

09 / ORCHESTRATION

Frontier judgment at the top. Economical execution underneath.

This architecture combines the strongest observed product and model behaviors without pretending that every task needs a committee.



Evidence before impressions.

Every conclusion traces back to a named baseline, frozen prompt, patch, verification record, timing, and explicitly labeled cost field.

ASSISTANTS

Codex, Claude Code, OpenCode, Pi

EVIDENCE HARNESS

Git worktrees, frozen prompts, JSON manifests, saved patches

FRONTIER MODELS

GPT-5.6 Sol and Terra; Claude Opus, Sonnet, Fable

VERIFICATION

Visible tests, private suites, browser QA, scope checks

OPEN MODELS

Kimi K2.7 and K3; Qwen3.8-27B; Qwen3-Coder-Next; DeepSeek V4; GLM 5.2 and 5.3

PORTABILITY

AGENTS.md, skills, read-only MCP, repository scripts

MODEL ACCESS

OpenRouter, Ollama Cloud, subscription routes

SHOWCASE

Next-compatible app, Vinext build, synthetic trade workspace

CONCLUSION

Build a team of roles, not a leaderboard of brands.

Codex is the default lead. Claude is the independent reviewer. OpenCode is the flexible worker harness. Pi is the neutral lab. Sol plans; Qwen executes cheaply; GLM handles hard asynchronous work; Fable buys speed; Kimi stays guarded until completion is more reliable. The open-weight trial reached 10/10 on the independent gate at \$0.46, while its candidate suite exposed a test-contract failure.

READ WITH CARE

- Most Phase 1 routes ran once; only Phase 2 T10 was repeated.
- Synthetic tasks show observed behavior, not universal rank.
- OpenRouter balance did not reconcile to saved Pi cost telemetry.
- The trade-capture workspace is a synthetic mock, not production financial software.
- AR8 is one open-weight observation; its candidate suite failed, so acceptance relies on the external evaluator.

11 / EVIDENCE INDEX

The claims are inspectable.

The website and this paper summarize repository evidence. These primary artifacts preserve the detailed narrative, run blocks, validation rules, and reproducibility record.

Phase 1 comprehensive report

01

[benchmark/reports/phase-1/PHASE-1-COMPREHENSIVE.md](#)

Assistant differentiation, benchmark tasks, distributed workflow, trade-capture artifact, and limitations.

Phase 2 comprehensive report

02

[benchmark/reports/phase-2/PHASE-2-COMPREHENSIVE.md](#)

Neutral Pi comparison, task-level outcomes, role recommendations, cost caveat, and audit findings.

Phase 2 result block

03

[benchmark/blocks/phase-2-pi-model-comparison-2026-08-23.results.json](#)

Structured scores, acceptance counts, elapsed time, telemetry labels, and status fields.

Neutral protocol

04

[benchmark/phase-2/PROTOCOL.md](#)

Fixed control surface, early-edit rule, timeouts, zero intervention, private checks, and saved-patch policy.

Final study report

05

[benchmark/reports/final-study-report.md](#)

Cross-phase conclusions, recommended operating model, and decision guidance.

Browser QA report

06

[benchmark/reports/results-site-browser-qa.md](#)

Visible validation record for the results interface and user-facing research experience.

SCOPE NOTE

The study records observed behavior on named synthetic tasks. It is a decision aid, not a universal model ranking or production-finance certification.

Benchmarks and methods that shaped the study.

The paper uses these public benchmark and evaluation threads for positioning; the repository artifacts remain the primary evidence for every reported result.

01

SWE-bench

Jimenez et al. (2024). SWE-bench: Can Language Models Resolve Real-World GitHub Issues? ICLR.

<https://www.swebench.com/>

02

SWE-Lancer

SWE-Lancer (2025). A software-engineering task benchmark for frontier language models.

<https://arxiv.org/abs/2502.12115>

03

Terminal-Bench

Terminal-Bench (2025). A benchmark for agents operating through terminal tools.

<https://www.tbench.ai/>

04

RE-Bench

RE-Bench (2024). A benchmark for research and engineering trajectories.

<https://metr.org/>

05

UTBoost

UTBoost (2025). Evidence on evaluator quality and coding-agent benchmark reliability.

<https://arxiv.org/abs/2506.09289>

REPRODUCIBILITY NOTE

All claims in this paper should be read alongside the versioned prompts, manifests, patches, evaluator records, and audit reports listed on the preceding evidence-index page.